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k-NN and biological similarity

Week 8: What "similar" means, and why the representation decides it.

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By the end of this week you should be able to

Learning outcomes served: LO2, LO3, LO5

- Implement k-NN classification and regression, and explain the role of k.
- Choose an appropriate distance metric, including for mixed-type biological data.
- Explain why k-NN requires feature scaling and demonstrate the effect.
- Explain how irrelevant features degrade nearest-neighbour search.
- Compare hand-engineered and learned representations under an identical k-NN classifier and interpret the difference.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- A great deal of biology is reasoning by similarity. BLAST finds homologues by sequence similarity; a clinician reasons from comparable past cases; a nutritionist groups foods by composition.
- The k-nearest-neighbour classifier makes that reasoning explicit, and in doing so exposes the question everything in this module turns on: similar in what space?

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The method

Part 2 · Concepts, then the mathematics

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Instance-based learning

No model is fitted at training time; the training data is the model. Cost is deferred to prediction, which makes k-NN a lazy learner.

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Choosing k

Small k gives a flexible, noise-sensitive boundary; large k smooths it and eventually predicts the majority class everywhere. On WDBC, cross-validated accuracy peaks around k = 3 to 9 and then declines.

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Key concepts

Euclidean distance Straight-line distance. Sensible for continuous features on comparable scales.

Manhattan distance Sum of absolute differences. Often more robust in high dimensions and when features are not directly comparable.

Hamming distance Number of positions at which two sequences differ. The natural metric for equal-length categorical or sequence data.

Heterogeneous distance Real biological records mix continuous, ordinal and categorical fields, so the metric must be composed per-field and the components weighted deliberately.

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Key concepts (continued)

Scaling is not optional Measured on WDBC: standardised k-NN at k = 5 reaches 96.3 per cent cross-validated accuracy, unstandardised 93.5 per cent. Nuclear area, in the hundreds, otherwise drowns out smoothness, around 0.1.

Irrelevant features Every uninformative dimension adds distance noise. This is why k-NN benefits so directly from a good representation, and why it is the fairest probe of representation quality.

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Distance metrics

$$\text{Euclidean} \quad \sqrt{\Sigma(a_i - b_i)^2}$$
$$\text{Manhattan} \quad \Sigma|a_i - b_i|$$
$$\text{Hamming} \quad \Sigma\, 1[a_i \neq b_i]$$

The metric defines what 'similar' means, and therefore defines the model. Real biological records mix continuous, ordinal and categorical fields, so the metric must be composed per field and the components weighted deliberately.

Euclidean continuous features on comparable scales

Manhattan often more robust in high dimensions

Hamming equal-length sequences or categorical data

cosine $1 - (a \cdot b)/(\|a\|\|b\|)$; usual choice for embeddings

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Distance-weighted voting

$$\text{vote}(c) = \sum \frac{1}{d(q, x_i)^2} \quad \text{over neighbours } i \text{ of class } c$$

Closer neighbours count for more, which softens the effect of choosing k slightly wrong and handles ties sensibly.

$d(q,x_i)$ distance from the query to neighbour i

k still needs choosing, but the result is less sensitive to it

lazy no model is fitted at training time; the data is the model, and all cost is deferred to prediction

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What k-NN is sensitive to, measured on WDBC

Nuclear area runs into the thousands and smoothness sits around 0.1, so without scaling the distance is essentially the difference in area.

Representation	Dimensions	Accuracy at k = 5
30 features, standardised	30	96.3%
30 features, NOT standardised	30	93.5%
Standardised + 60 pure-noise features	90	92.1%

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Why k-NN suffers from irrelevant features and a tree does not

The Week 7 comparison is run again here with the classifier held constant, which is a clean experimental design.

- k-NN collapses every feature into a single distance, so every irrelevant dimension adds noise to that distance
- A decision tree makes a per-feature decision and can simply never split on a useless column
- Small k gives a flexible, noise-sensitive boundary
- Large k smooths it, and as k approaches n the model predicts the majority class everywhere
- This is exactly why k-NN is the fairest probe of representation quality: hold the classifier fixed and change only what it measures distance in

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Worked example

Part 3 · Two representations, one classifier

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Two representations, one classifier

- Hold the classifier fixed at k-NN with k = 5 and change only what it measures distance in. Any difference in accuracy is attributable to the representation alone.
- That is a clean experimental design and it is why this comparison belongs with k-NN rather than with a more powerful model that could compensate.
- On the family task, the hand-engineered representation of amino-acid composition plus four biophysical descriptors reaches essentially perfect accuracy, because that is precisely how the families differ. The learned embedding does respectably but no better.
- Buying a foundation model here would be spending compute to lose accuracy.

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Two representations, one classifier (continued)

- On the ordered-triad task the picture inverts. Composition is identical between the classes by construction, so the engineered representation sits at chance, around 0.50. The learned representation does clearly better. The classifier never changed; only what it could see did.

- One further observation worth drawing out: the embedding's k-NN accuracy on the family task is lower than its logistic-regression accuracy on the same data. In high dimensions, many low-variance directions contribute noise to the distance while a linear model can simply give them small weights.

- That is the irrelevant-feature problem, and a bigger embedding does not automatically make it better.

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A better representation gives better neighbours only when the thing it represents better is the thing you are predicting.

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Try this interactively

Biological Similarity Explorer

Is it the classifier or the representation that decides who the neighbours are?

`weeks/week08.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · Embeddings plus a classical head

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Embeddings plus a classical head

- Embeddings followed by k-NN or logistic regression is the most useful foundation-model recipe in practice, and it is the one that gets underused.
- It needs no gradient steps through the large model, runs on a laptop once the embeddings are computed, keeps the downstream model small enough to interpret and to cross-validate honestly, and works with a few hundred labels.
- Before you consider fine-tuning in Week 11, this is the baseline that fine-tuning has to beat.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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Your k-NN on foundation-model embeddings outperforms k-NN on your engineered features. Give two explanations other than "the embedding captures more biology", and say what evidence would let you distinguish them.

Think like a scientist. Bring your answer to the laboratory.

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What counts as similar?

This week's laboratory · Establish experimentally that the representation, not the classifier, determines nearest-neighbour quality. · 2 hours

- Fit k-NN on WDBC with and without StandardScaler, at k = 1, 3, 5, 9, 15, 31 and 51. Produce one plot with both curves.
- Explain the gap between the curves by inspecting the feature ranges. Which feature dominates the unscaled distance, and by what factor?
- Compare Euclidean, Manhattan and Chebyshev distance at the best k. Is the difference larger or smaller than the scaling effect?
- Add 50 columns of pure Gaussian noise to the standardised data and refit. Quantify how much accuracy you lost, then repeat with 200 noise columns.
- Repeat task 4 with a decision tree instead of k-NN. Explain why the two models degrade so differently.
- On the protein set, hold k-NN fixed and vary only the representation, for both labels. Report accuracy, macro-F1 and the confusion matrix for each of the four combinations.
- Answer in writing: does a better representation always produce better nearest neighbours? Use your own numbers as evidence.

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Self-check

- Increasing k from 1 to 51 on a 569-sample dataset. What happens at large k?
- A) Accuracy keeps improving
- B) The decision boundary smooths and eventually the model approaches predicting the majority class everywhere
- C) The model overfits more
- D) Nothing changes

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Self-check

- Why does k-NN suffer more than a decision tree from irrelevant features?
- A) It uses more memory
- B) It collapses all features into a single distance, so every irrelevant dimension adds noise to that distance; a tree can simply never split on the feature
- C) It cannot handle continuous features
- D) It uses the wrong metric

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References and further reading

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Lin, Z. et al. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. Science 379:1123-1130. doi:10.1126/science.ade2574

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Questions

Next week: continue to Week 9.

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